

## Names, data checks, and new variables

## Placeholder names

| Name             | Stands for                             |
|------------------|----------------------------------------|
| dat, analysis    | data frames                            |
| y                | outcome                                |
| x, x1, x2, x3, z | numerical predictors                   |
| group            | categorical predictor                  |
| m, m_multi, ...  | fitted models                          |
| n, k, G          | number of observations, slopes, groups |

## Read and inspect the data

```
dat <- read.csv("data.csv")

dim(dat)           # number of rows and columns
nrow(dat)          # number of rows
names(dat)         # variable names
head(dat)          # first six rows
summary(dat)       # summary of each variable
table(dat$group)   # count per category
mean(dat$y)        # mean
sd(dat$y)          # standard deviation
cor(dat$x, dat$y)  # correlation
tapply(dat$y, dat$group, mean) # mean of y per category
```

## New variables and the analysis sample

```
dat$x_1k <- dat$x_euros / 1000 # in 1,000 euros
dat$group <- factor(dat$group) # categorical
dat$d <- as.integer(dat$group == "A") # 1 if A, else 0

# rows that meet the conditions
analysis <- subset(dat, x > 0 & group == "A")
```

## Lecture 1: Simple linear regression

## Population model and fitted line

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i.$$

## OLS estimates in simple regression

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{s_{XY}}{s_X^2} = r_{XY} \frac{s_Y}{s_X}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

## Residual and sum of squared residuals

$$e_i = Y_i - \hat{Y}_i, \quad \text{SSE} = \sum_{i=1}^n e_i^2.$$

## Change of units by a factor c

| Multiplied by c       | New intercept     | New slope           |
|-----------------------|-------------------|---------------------|
| outcome only          | $c \hat{\beta}_0$ | $c \hat{\beta}_1$   |
| predictor only        | $\hat{\beta}_0$   | $\hat{\beta}_1 / c$ |
| outcome and predictor | $c \hat{\beta}_0$ | $\hat{\beta}_1$     |

## Fit a simple regression

```
m <- lm(y ~ x, data = dat)
coef(m)           # estimates
summary(m)        # coefficient table and fit
fitted(m)         # fitted values
residuals(m)      # residuals
```

## Scatter plot with the fitted line

```
plot(y ~ x, data = dat,
     xlab = "Predictor (unit)", ylab = "Outcome (unit)")
abline(m, col = "blue", lwd = 2)
```

## Fitted values for chosen values x\_A and x\_B of the predictor

```
b <- coef(m)
b[1] + b[2] * x_A # fitted value by hand

new_cases <- data.frame(x = c(x_A, x_B))
predict(m, newdata = new_cases) # fitted values
```

## Lecture 2: What the model assumes and how residuals reveal it

## Residual plot and normal Q-Q plot

```
plot(fitted(m), residuals(m),
     xlab = "Fitted value", ylab = "Residual")
abline(h = 0, lty = 2)
```

```
qqnorm(rstandard(m))
qqline(rstandard(m))
```

## Standardised residuals and Cook's distance

```
summary(residuals(m)) # summary of the residuals
rstandard(m)          # standardised residuals
cooks.distance(m)     # Cook's distances
nobs(m)               # number of observations n
```

## Flagged observations

$$|\text{standardised residual}| > 3, \quad \text{Cook's distance} > \frac{4}{n}.$$

```
sum(abs(rstandard(m)) > 3) # how many beyond ±3
sum(cooks.distance(m) > 4 / nobs(m)) # how many above 4/n
```

## Lecture 3: Coefficient inference, fit, and the F-test

## Two-sided test of one coefficient

$$H_0 : \beta_j = 0, \quad H_1 : \beta_j \neq 0, \quad t = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}.$$

Reject  $H_0$  at the 5% level if  $|t_{\text{obs}}| \geq t_{0.975, df}$ .

## Degrees of freedom and 95% confidence interval, model with k slopes and an intercept

$$df = n - k - 1, \quad \hat{\beta}_j \pm t_{0.975, df} \text{SE}(\hat{\beta}_j),$$

simple regression:  $k = 1$ ,  $df = n - 2$ .

```
summary(m)           # coefficient table and fit
confint(m, level = 0.95) # 95% confidence intervals
df.residual(m)        # residual degrees of freedom
qt(0.975, df.residual(m)) # critical value, 95% interval
```

## Labels in the output of summary(m)

| Label in the output     | Symbol                                                    |
|-------------------------|-----------------------------------------------------------|
| Estimate                | $\hat{\beta}_j$                                           |
| Std. Error              | $\text{SE}(\hat{\beta}_j)$                                |
| t value                 | $t$                                                       |
| Pr(> t )                | two-sided p-value for $H_0 : \beta_j = 0$                 |
| Residual standard error | $s$ , with $df = n - k - 1$                               |
| Multiple R-squared      | $R^2$                                                     |
| Adjusted R-squared      | $R_{\text{adjusted}}^2$                                   |
| F-statistic             | overall $F$ , with $k$ and $n - k - 1$ degrees of freedom |

## The t statistic and the two-sided p-value step by step

```
tab <- summary(m)$coefficients
t_value <- tab["x", "Estimate"] / tab["x", "Std. Error"]
p_value <- 2 * pt(-abs(t_value), df = df.residual(m))
```

Sums of squares and  $R^2$ 

$$\text{SST} = \sum_i (y_i - \bar{y})^2, \quad \text{SSE} = \sum_i (y_i - \hat{y}_i)^2,$$

$$\text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2,$$

$$\text{SST} = \text{SSR} + \text{SSE}, \quad R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{\text{SSR}}{\text{SST}}.$$

```
y <- dat$y # outcome values
yhat <- fitted(m) # fitted values
```

```
SST <- sum((y - mean(y))^2)
SSE <- sum(residuals(m)^2)
SSR <- SST - SSE
R2 <- 1 - SSE / SST
```

**Residual standard error and adjusted  $R^2$** 

$$s = \sqrt{\frac{\text{SSE}}{n - k - 1}}, \quad R^2_{\text{adjusted}} = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}.$$

```
summary(m)$r.squared      # R squared
summary(m)$fstatistic     # overall F, k, and n - k - 1
```

**Overall F-test, model with  $k$  slopes**

$$H_0 : \beta_1 = \dots = \beta_k = 0, \\ H_1 : \text{at least one of } \beta_1, \dots, \beta_k \text{ is not } 0,$$

$$F = \frac{\text{SSR}/k}{\text{SSE}/(n - k - 1)}, \quad \text{simple regression: } F = t^2.$$

**Robust standard errors (HC1)**

```
library(lmtest)
library(sandwich)

# robust standard errors, t statistics, and p-values
coeftest(m, vcov. = vcovHC(m, type = "HC1"))
# robust 95% confidence intervals
coefci(m, vcov. = vcovHC(m, type = "HC1"))
```

**Lecture 4: Multiple regression and what controls do****Model with  $k$  predictors**

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \varepsilon_i.$$

```
m_multi <- lm(y ~ x1 + x2 + x3, data = analysis)
summary(m_multi)      # coefficient table and fit
confint(m_multi)      # 95% confidence intervals
```

**Fitted difference for changes  $\Delta x_1, \dots, \Delta x_k$  of the predictors**

$$\Delta \hat{Y} = \hat{\beta}_1 \Delta x_1 + \dots + \hat{\beta}_k \Delta x_k.$$

**Fitted difference between two cases A and B**

```
cases <- data.frame(
  x1 = c(x1_A, x1_B),
  x2 = c(x2_A, x2_B),
  x3 = c(x3_A, x3_B)
)
fits <- predict(m_multi, newdata = cases) # fitted values
fits[2] - fits[1]                        # B minus A
```

**Regression table with one column per model**

```
# install once: install.packages("modelsummary")
library(modelsummary)
m_a <- lm(y ~ x1, data = analysis)
m_b <- lm(y ~ x1 + x2, data = analysis)
modelsummary(
  list("(1)" = m_a, "(2)" = m_b, "(3)" = m_multi),
  stars = c("*" = 0.05, "***" = 0.01, "****" = 0.001),
  gof_map = c("nobs", "r.squared", "adj.r.squared")
)
```

**Lecture 5: Categories as predictors and joint tests****Reference category and model with a categorical predictor**

```
analysis$group <- factor(analysis$group)
# categories: the first one is the reference
levels(analysis$group)
analysis$group <- relevel(analysis$group,
  ref = "reference")

m_group <- lm(y ~ x1 + x2 + group, data = analysis)
summary(m_group)
coef(m_group)
confint(m_group)
```

**The same model with another reference category**

```
analysis$group_alt <- relevel(analysis$group,
  ref = "other")
m_alt <- lm(y ~ x1 + x2 + group_alt, data = analysis)
# TRUE if the fitted values of the two models are equal
all.equal(fitted(m_group), fitted(m_alt))
```

**Partial F-test of nested models**

```
reduced <- lm(y ~ x1 + x2, data = analysis)
complete <- lm(y ~ x1 + x2 + group, data = analysis)
anova(reduced, complete)
```

Complete model:  $k_C$  slopes. Reduced model: the complete model without  $q$  of its terms, with the same outcome and the same observations.

$H_0$  : the  $q$  coefficients left out of the reduced model are all 0,

$$F = \frac{(\text{SSE}_R - \text{SSE}_C)/q}{\text{SSE}_C/(n - k_C - 1)}.$$

**Labels in the output of `anova(reduced, complete)`**

| Label in the output | Symbol                                           |
|---------------------|--------------------------------------------------|
| Res.Df              | $n - k - 1$ of each model                        |
| RSS                 | $\text{SSE}_R$ in row 1, $\text{SSE}_C$ in row 2 |
| Df                  | $q$                                              |
| Sum of Sq           | $\text{SSE}_R - \text{SSE}_C$                    |
| F                   | partial $F$                                      |
| Pr(>F)              | p-value of the partial F-test                    |

**One-way ANOVA: one model,  $G$  groups**

```
m_anova <- lm(y ~ group, data = analysis)
anova(m_anova) # ANOVA table
```

$$H_0 : \text{all } G \text{ group means are equal, } F = \frac{\text{SSR}/(G - 1)}{\text{SSE}/(n - G)}.$$

**Labels in the output of `anova(m_anova)`**

| Label in the output | Symbol                               |
|---------------------|--------------------------------------|
| Df                  | $G - 1$ (group), $n - G$ (Residuals) |
| Sum Sq              | SSR (group), SSE (Residuals)         |
| Mean Sq             | Sum Sq / Df                          |
| F value             | $F$                                  |
| Pr(>F)              | p-value of the F-test                |

**Lecture 6: Interactions and curvature****Interaction of a numerical predictor  $X$  with a dummy  $D$** 

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 D + \hat{\beta}_3 XD.$$

| Group   | Intercept                       | Slope                           |
|---------|---------------------------------|---------------------------------|
| $D = 0$ | $\hat{\beta}_0$                 | $\hat{\beta}_1$                 |
| $D = 1$ | $\hat{\beta}_0 + \hat{\beta}_2$ | $\hat{\beta}_1 + \hat{\beta}_3$ |

```
m_interaction <- lm(y ~ x * group + z, data = analysis)
# x * group expands to x + group + x:group
summary(m_interaction)
```

**Centred predictor**

```
# x minus its mean
analysis$x_c <- analysis$x - mean(analysis$x)
m_centred <- lm(y ~ x_c * group + z, data = analysis)
```

**Quadratic model**

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2.$$

Fitted change from  $x$  to  $x + 1$ :  $\hat{\beta}_1 + \hat{\beta}_2(2x + 1)$ .

```
m_linear <- lm(y ~ x + z, data = analysis)
m_quadratic <- lm(y ~ x + I(x^2) + z, data = analysis)
# I(x^2) is the square of x

summary(m_quadratic)
anova(m_linear, m_quadratic)
```

## Commands by purpose

| Purpose                              | R command                                                 |
|--------------------------------------|-----------------------------------------------------------|
| Fit a model                          | <code>lm(y ~ predictors, data = dat)</code>               |
| Full model output                    | <code>summary(m)</code>                                   |
| Coefficients only                    | <code>coef(m)</code>                                      |
| 95% coefficient intervals            | <code>confint(m)</code>                                   |
| Fitted values and residuals          | <code>fitted(m), residuals(m)</code>                      |
| Summary of the residuals             | <code>summary(residuals(m))</code>                        |
| Standardised residuals               | <code>rstandard(m)</code>                                 |
| Cook's distance                      | <code>cooks.distance(m)</code>                            |
| Fitted values for new cases          | <code>predict(m, newdata = ...)</code>                    |
| $R^2$ , adjusted $R^2$ , overall F   | <code>summary(m)</code>                                   |
| Residual degrees of freedom          | <code>df.residual(m)</code>                               |
| Critical value of a 95% interval     | <code>qt(0.975, df.residual(m))</code>                    |
| Partial F-test of nested models      | <code>anova(reduced, complete)</code>                     |
| ANOVA table of one model             | <code>anova(m)</code>                                     |
| Reference category of a factor       | <code>relevel(factor(group), ref = "category")</code>     |
| Interaction                          | <code>x * group</code>                                    |
| Square in a formula                  | <code>I(x^2)</code>                                       |
| Robust standard errors               | <code>coeftest(m, vcov. = vcovHC(m, type = "HC1"))</code> |
| Regression table with several models | <code>modelsummary(list(...))</code>                      |